

LEC14 - Light & Matter

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Learning Goals

- Describe the interactions between light and matter, including emission, reflection, absorption, and scattering
- Explain the Doppler shift and its effect on stellar spectra

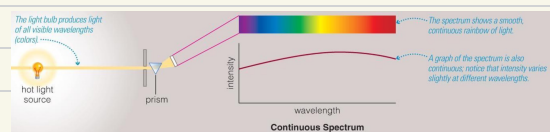
Term Test:

- 50 minutes allocated, so arrive early if possible
- Only bring pen/pencil & T-Card
- No calculators needed or allowed
- Other belongings must be left in front of the room
- Phones and smartwatches away & silenced
- Formulas will be given if needed

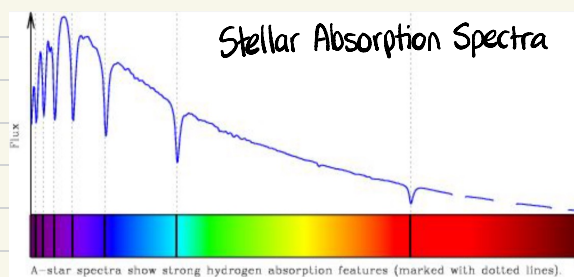
Continuum: A continuous sequence in which adjacent elements are not very different from each other. However, extremes can be quite distinct

Kinds of Spectra - Thermal Continuum Spectrum (Blackbody):

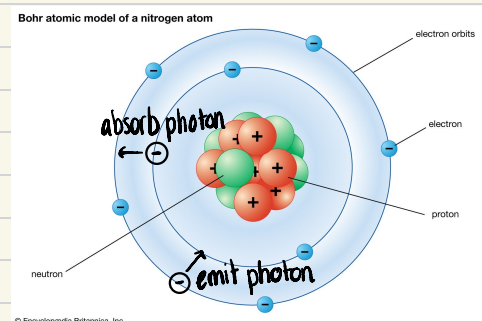
- Thermal emission** is determined by the temperature of the object. It is a **continuum** of light across all wavelengths
- Thermal emission is governed by 2 laws
 - The curve for a hotter object is everywhere above the curve for a cooler object, showing that hotter objects emit more radiation per unit surface area at every wavelength
 - The **peak wavelength** is further to the left for hotter objects, showing that hotter objects emit more of their light at shorter wavelengths

Spectra of Stars:

- Approximately thermal spectra, but with spectral absorption lines

Matter - Atoms:

- Bohr's model of the atom states that electrons can only be in **one** of numerous orbits
 - Oversimplified model, but sufficient enough for this course
- Energy levels represent the amount of Kinetic Energy and Electromagnetic Potential Energy the electron has
- Transitions require **exactly** the right amount of energy
- If a photon with the right amount of energy "arrives", it can excite a transition

Photons and Energy:

- Light also behaves as a particle called a photon
- Each photon carries a fixed amount of energy
 - Proportional to frequency
- $E = h \cdot f$
 - $E = h \cdot \frac{c}{\lambda}$

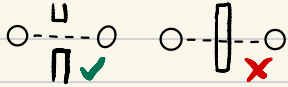
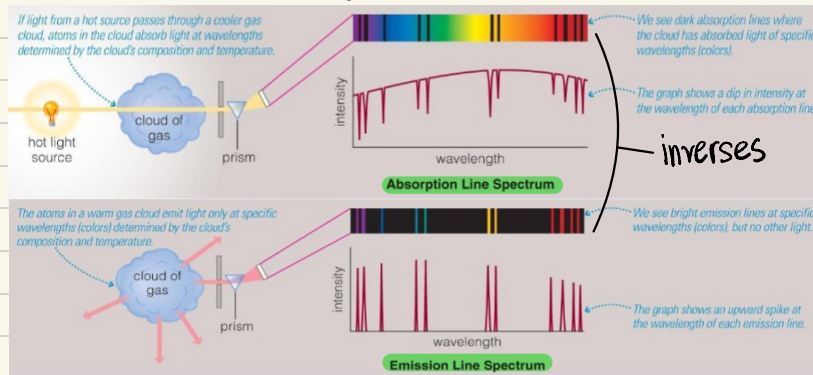
- Thermal emission is governed by 2 laws. The first is that hotter objects emit more radiation. The second is that hotter objects emit shorter wavelengths of light (more blue & purple)
- Bohr's model of the atom states that electrons can only be in a single distinct orbit at a time. Electrons can absorb photons to move to a higher energy level, and emit a photon to jump to a lower energy orbit.

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Line-of-sight Motion:

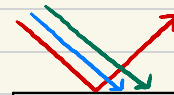
Waves can only travel in a direct visual path from the source to the receiver without obstacles

Transitions between Energy Levels:Emission Line Spectra

- In a warm gas, the higher temperature is caused by a higher average kinetic energy
 - This means more collisions "kicking" electrons into higher energy levels
 - These then decay back into their lower (and more stable) energy levels
 - Electrons emit a characteristic photon when going down a level

Spectra:

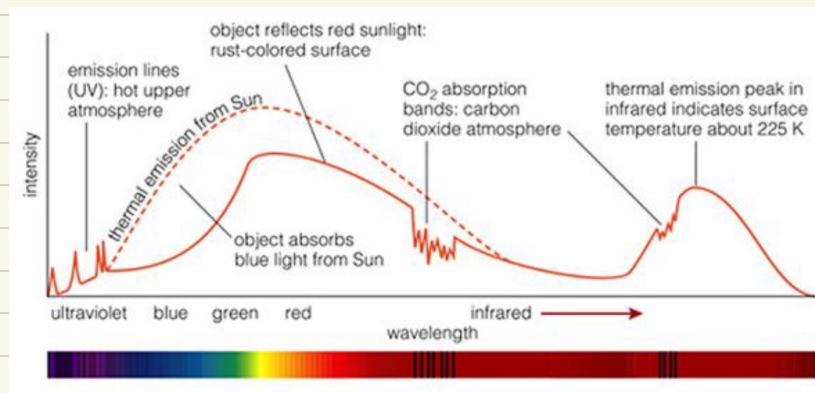
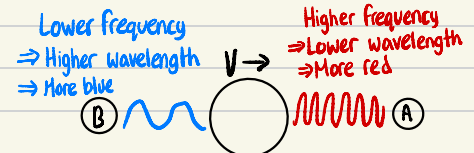
- Spectra are the "fingerprints" of elements
 - Each element has a unique emission spectra
- Stellar spectra look like thermal continuum, but with absorption lines due to cooler gas in the atmosphere of the star
- Matter can absorb and reflect light
 - The reason an object might look red is because it absorbs blue and green light, while reflecting red light into our eyes



Red object (apple etc.)

The Doppler Effect:

- Mostly known in sound waves
 - Siren sounds high pitched when approaching, but low pitch when moving away
- Light waves also undergo Doppler shifts
 - Redshift = Object moving away from us
 - Rest Frame = Object is at rest relative to us
 - Blueshift = Object moving towards us
- Doppler shifts only measure line-of-sight motion
- Doppler shifts allow us to measure orbits and discover planets



- Emission spectra can uniquely identify an element
- The Doppler effect also affects light waves, causing a redshift if the object is moving towards the observer, and a blueshift if the object is moving away from the observer